

EDEXCEL INTERNATIONAL A LEVEL

WMA11/01 May/June 2024 Worked Solutions

May/June 2024 paper rebuilt with the worked solution after each question.

11

paper questions

WMA11

Pure 1

**Single-
paper
solutions**standalone IAL
route**Dr Eslam Ahmed**Prepared for Dr Eslam Ahmed - eliteigcse.com**P1**

PAST PAPER

WMA11/01 May/June 2024

May/June 2024 | 11 questions | 75 marks

11

questions

75

marks

Answers

worked solution
after each
question

Question 1

Integration

1. Find

$$\int \left(10x^4 - \frac{3}{2x^3} - 7 \right) dx$$

giving each term in simplest form.

(3)

(Total for Question 1 is 3 marks)

Worked Solution - Question 1

1. Rewrite the fraction

$$10x^4 - \frac{3}{2x^3} - 7 = 10x^4 - \frac{3}{2}x^{-3} - 7.$$

2. Integrate term by term

$$\int 10x^4 dx = 2x^5, \int -\frac{3}{2}x^{-3} dx = \frac{3}{4}x^{-2}, \text{ and } \int -7 dx = -7x.$$

3. Write in simplest form

$$2x^5 + \frac{3}{4x^2} - 7x + c.$$

Final answer

$$2x^5 + \frac{3}{4x^2} - 7x + c.$$

Question 2

Indices & Surds

2. (i) Given that $m = 2^n$, express each of the following in simplest form in terms of m .

(a) 2^{n+3}

(1)

(b) 16^{3n}

(2)

- (ii) **In this question you must show all stages of your working.**

Solutions relying on calculator technology are not acceptable.

Solve the equation

$$x\sqrt{3} - 3 = x + \sqrt{3}$$

giving your answer in the form $p + q\sqrt{3}$ where p and q are integers.

(3)

(Total for Question 2 is 6 marks)

Worked Solution - Question 2

Primary topic

1. Use m equals 2 to the n

$$2^{n+3} = 2^n \times 2^3 = 8m.$$

2. Rewrite 16 as a power of 2

$$16^{3n} = (2^4)^{3n} = 2^{12n} = (2^n)^{12} = m^{12}.$$

3. Collect the x terms

$$x\sqrt{3} - 3 = x + \sqrt{3} \text{ gives } x(\sqrt{3} - 1) = 3 + \sqrt{3}.$$

4. Rationalise

$$x = \frac{3 + \sqrt{3}}{\sqrt{3} - 1} \times \frac{\sqrt{3} + 1}{\sqrt{3} + 1} = \frac{(3 + \sqrt{3})(\sqrt{3} + 1)}{2}.$$

5. Simplify

$$x = \frac{4\sqrt{3} + 6}{2} = 3 + 2\sqrt{3}.$$

Final answer

$$(i)(a) 8m. \quad (i)(b) m^{12}. \quad (ii) x = 3 + 2\sqrt{3}.$$

Question 3

Transformations

3.

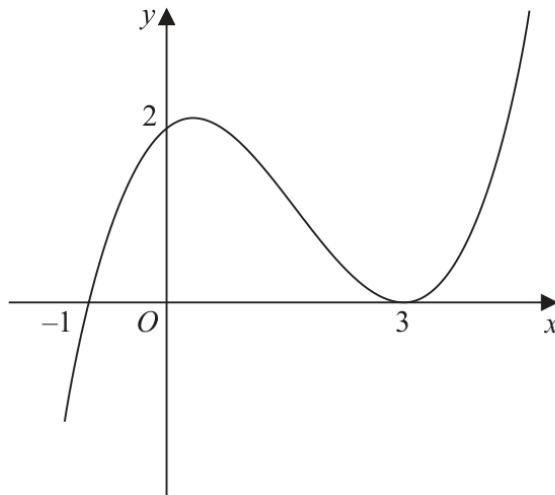


Figure 1

Figure 1 shows a sketch of the curve with equation $y = f(x)$.

The curve passes through the points $(-1, 0)$ and $(0, 2)$ and touches the x -axis at the point $(3, 0)$.

On separate diagrams, sketch the curve with equation

(a) $y = f(x + 3)$ (3)

(b) $y = f(-3x)$ (3)

On each diagram, show clearly the coordinates of all the points where the curve cuts or touches the coordinate axes.

(Total for Question 3 is 6 marks)

Worked Solution - Question 3

Primary topic

1. Read the original key points

The original curve passes through $(-1, 0)$ and $(0, 2)$, and touches the x-axis at $(3, 0)$.

2. Transform y equals f of x plus 3

$y = f(x + 3)$ is a translation 3 units left. So $(-1, 0)$ moves to $(-4, 0)$, $(0, 2)$ moves to $(-3, 2)$, and $(3, 0)$ moves to $(0, 0)$.

3. State the axis intercepts for part a

The curve cuts the x-axis at $(-4, 0)$ and touches it at $(0, 0)$. The y-intercept is also $(0, 0)$.

4. Transform y equals f of negative $3x$

For $y = f(-3x)$, a point with original x-coordinate X moves to $x = -\frac{X}{3}$.

5. State the axis intercepts for part b

$(-1, 0)$ moves to $(\frac{1}{3}, 0)$, $(3, 0)$ moves to $(-1, 0)$, and $(0, 2)$ stays at $(0, 2)$.

Final answer

(a) intercepts $(-4, 0)$ and $(0, 0)$, with $(0, 0)$ a touch point.

(b) intercepts $(-1, 0)$, $(\frac{1}{3}, 0)$ and $(0, 2)$, with $(-1, 0)$ a touch point.

Question 4

Quadratics

4. The curve C_1 has equation

$$y = x^2 + kx - 9$$

and the curve C_2 has equation

$$y = -3x^2 - 5x + k$$

where k is a constant.

Given that C_1 and C_2 meet at a single point P

- (a) show that

$$k^2 + 26k + 169 = 0 \quad (3)$$

- (b) Hence find the coordinates of P

(3)

(Total for Question 4 is 6 marks)

Worked Solution - Question 4

Primary topic

1. Equate the curves

$$x^2 + kx - 9 = -3x^2 - 5x + k \text{ gives } 4x^2 + (k + 5)x - (k + 9) = 0.$$

2. Use one intersection only

For a single point of intersection, the discriminant is zero.

3. Form the condition on k

$$(k + 5)^2 - 4(4)(-(k + 9)) = 0, \text{ so } k^2 + 26k + 169 = 0.$$

4. Find k

$$k^2 + 26k + 169 = (k + 13)^2, \text{ so } k = -13.$$

5. Find x

$$\text{Substituting } k = -13 \text{ gives } 4x^2 - 8x + 4 = 0, \text{ so } 4(x - 1)^2 = 0 \text{ and } x = 1.$$

6. Find y

$$y = x^2 + kx - 9 = 1 - 13 - 9 = -21, \text{ so } P = (1, -21).$$

Final answer

$$P = (1, -21).$$

Question 5

Radians

5.

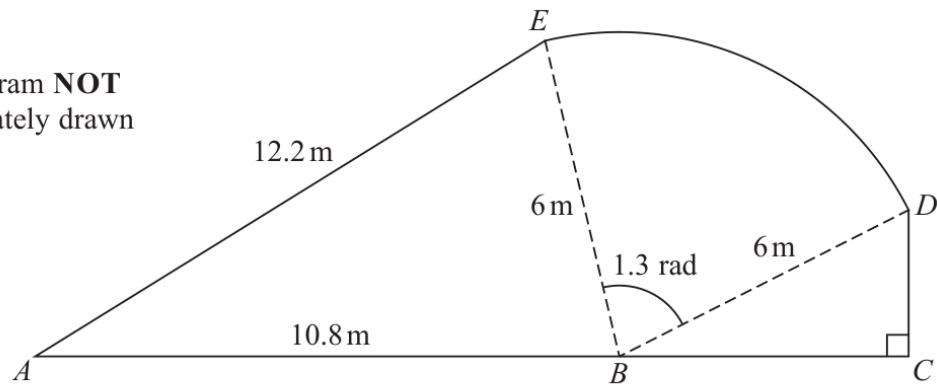
Diagram **NOT**
accurately drawn

Figure 2

Figure 2 shows the plan view of a garden.

The shape of the garden $ABCDEA$ consists of a triangle ABE and a right-angled triangle BCD joined to a sector BDE of a circle with radius 6 m and centre B .

The points A , B and C lie on a straight line with $AB = 10.8$ m

Angle $BCD = \frac{\pi}{2}$ radians, angle $EBD = 1.3$ radians and $AE = 12.2$ m

- (a) Find the area of the sector BDE , giving your answer in m^2 (2)
- (b) Find the size of angle ABE , giving your answer in radians to 2 decimal places. (2)
- (c) Find the area of the garden, giving your answer in m^2 to 3 significant figures. (3)

(Total for Question 5 is 7 marks)

Worked Solution - Question 5

Primary topic

1. Find the sector area

$$\text{Area of sector } BDE = \frac{1}{2}(6^2)(1.3) = 23.4 \text{ m}^2.$$

2. Use the cosine rule in triangle ABE

$$12.2^2 = 10.8^2 + 6^2 - 2(10.8)(6) \cos \angle ABE.$$

3. Find angle ABE

$$\cos \angle ABE = \frac{10.8^2 + 6^2 - 12.2^2}{2(10.8)(6)}, \text{ so } \angle ABE \approx 1.54 \text{ radians.}$$

4. Find the area of triangle ABE

$$\text{Area } ABE = \frac{1}{2}(10.8)(6) \sin(1.54147\dots) \approx 32.386.$$

5. Find angle DBC

$$\angle DBC = \pi - 1.3 - \angle ABE \approx 0.30012 \text{ radians.}$$

6. Find the area of triangle BCD

$$\text{Area } BCD = \frac{1}{2}(6 \cos \angle DBC)(6 \sin \angle DBC) \approx 5.084.$$

7. Add the three areas

$$23.4 + 32.386 + 5.084 \approx 60.9 \text{ m}^2 \text{ to 3 significant figures.}$$

Final answer

(a) 23.4 m^2 .

(b) 1.54 radians.

(c) 60.9 m^2 .

Question 6

Inequalities

6.

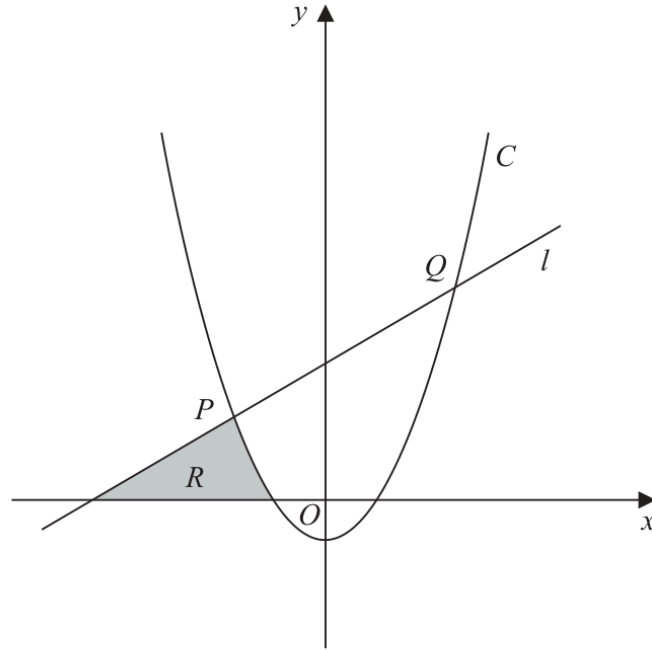


Figure 3

In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

Figure 3 shows

- the line l with equation $y - 5x = 75$
- the curve C with equation $y = 2x^2 + x - 21$

The line l intersects the curve C at the points P and Q , as shown in Figure 3.

(a) Find, using algebra, the coordinates of P and the coordinates of Q .

(4)

The region R , shown shaded in Figure 3, is bounded by C , l and the x -axis.

(b) Use inequalities to define the region R .

(3)

(Total for Question 6 is 7 marks)

Worked Solution - Question 6

1. Write the line as y equals

$$y - 5x = 75, \text{ so } y = 5x + 75.$$

2. Equate line and curve

$$5x + 75 = 2x^2 + x - 21, \text{ so } 2x^2 - 4x - 96 = 0.$$

3. Solve the quadratic

$$x^2 - 2x - 48 = 0 = (x - 8)(x + 6), \text{ so } x = -6 \text{ or } x = 8.$$

4. Find the y -coordinates

Using $y = 5x + 75$, $x = -6$ gives $y = 45$ and $x = 8$ gives $y = 115$.

5. State P and Q

From the diagram, $P = (-6, 45)$ and $Q = (8, 115)$.

6. Use the x -axis and left branch

The shaded region is above the x -axis and on the left part of the diagram, so $y \geq 0$ and $x \leq 0$.

7. Use the line and curve as upper boundaries

The region is below both the line and the curve, so $y \leq 5x + 75$ and $y \leq 2x^2 + x - 21$.

Final answer

(a) $P = (-6, 45)$, $Q = (8, 115)$.

(b) $y \geq 0$, $y \leq 5x + 75$, $y \leq 2x^2 + x - 21$, $x \leq 0$.

Question 7

Differentiation

7. The curve C has equation $y = f(x)$ where

$$f(x) = 2x^3 - kx^2 + 14x + 24$$

and k is a constant.

- (a) Find, in simplest form,

(i) $f'(x)$

(ii) $f''(x)$

(3)

The curve with equation $y = f'(x)$ intersects the curve with equation $y = f''(x)$ at the points A and B .

Given that the x coordinate of A is 5

- (b) find the value of k .

(2)

- (c) Hence find the coordinates of B .

(3)

(Total for Question 7 is 8 marks)

Worked Solution - Question 7

1. Differentiate once

$$f'(x) = 6x^2 - 2kx + 14.$$

2. Differentiate again

$$f''(x) = 12x - 2k.$$

3. Use the intersection at A

At an intersection of $y = f'(x)$ and $y = f''(x)$, $f'(x) = f''(x)$. Given $x_A = 5$, substitute $x = 5$.

4. Find k

$$6(5)^2 - 2k(5) + 14 = 12(5) - 2k, \text{ so } 164 - 10k = 60 - 2k \text{ and } k = 13.$$

5. Find the other x-coordinate

With $k = 13$, $6x^2 - 26x + 14 = 12x - 26$, so $6x^2 - 38x + 40 = 0$. Hence $(3x - 4)(x - 5) = 0$.

6. Use the point B

The other point has $x = \frac{4}{3}$. Then $y = f''(\frac{4}{3}) = 12(\frac{4}{3}) - 26 = -10$.

Final answer

$$(a)(i) f'(x) = 6x^2 - 2kx + 14. \quad (a)(ii) f''(x) = 12x - 2k.$$

$$(b) k = 13.$$

$$(c) B = (\frac{4}{3}, -10).$$

Question 8

Graphs of Functions

8. The curve C_1 has equation

$$y = x(4 - x^2)$$

- (a) Sketch the graph of C_1 showing the coordinates of any points of intersection with the coordinate axes.

(3)

The curve C_2 has equation $y = \frac{A}{x}$ where A is a constant.

- (b) Show that the x coordinates of the points of intersection of C_1 and C_2 satisfy the equation

$$x^4 - 4x^2 + A = 0$$

(1)

- (c) Hence find the range of possible values of A for which C_1 meets C_2 at 4 distinct points.

(3)

(Total for Question 8 is 7 marks)

Worked Solution - Question 8

Primary topic

1. Find the intercepts of C1

$y = x(4 - x^2)$, so $y = 0$ when $x = 0$ or $x = \pm 2$. The curve crosses the axes at $(-2, 0)$, $(0, 0)$ and $(2, 0)$.

2. Use the cubic shape

$y = 4x - x^3$ has negative leading cubic term, so the left tail rises and the right tail falls.

3. Equate C1 and C2

$$x(4 - x^2) = \frac{A}{x}.$$

4. Multiply by x

$x^2(4 - x^2) = A$, so $4x^2 - x^4 = A$ and therefore $x^4 - 4x^2 + A = 0$.

5. Let u equal x squared

Let $u = x^2$. Then $u^2 - 4u + A = 0$.

6. Require two positive u-values

For 4 distinct x-values, the quadratic in u must have two distinct positive roots.

7. Find the range of A

The discriminant gives $16 - 4A > 0$, so $A < 4$. The product of the two roots is A , so $A > 0$. Therefore $0 < A < 4$.

Final answer

(a) x-intercepts $(-2, 0)$, $(0, 0)$, $(2, 0)$.

(b) $x^4 - 4x^2 + A = 0$.

(c) 0

Question 9

Straight Line

9. Given that

- the point A has coordinates $(4, 2)$
- the point B has coordinates $(15, 7)$
- the line l_1 passes through A and B

(a) find an equation for l_1 , giving your answer in the form $px + qy + r = 0$ where p , q and r are integers to be found.

(3)

The line l_2 passes through A and is parallel to the x -axis.

The point C lies on l_2 so that the length of BC is $5\sqrt{5}$

(b) Find both possible pairs of coordinates of the point C .

(4)

(c) Hence find the minimum possible area of triangle ABC .

(2)

(Total for Question 9 is 9 marks)

Worked Solution - Question 9

1. Find the gradient of l_1

$$m = \frac{7 - 2}{15 - 4} = \frac{5}{11}.$$

2. Use point A

$$y - 2 = \frac{5}{11}(x - 4).$$

3. Rearrange

$$11y - 22 = 5x - 20, \text{ so } 5x - 11y + 2 = 0.$$

4. Use the horizontal line l_2

Since l_2 passes through $A(4, 2)$ and is parallel to the x-axis, point C has coordinates $(c, 2)$.

5. Use BC equals 5 root 5

$$(c - 15)^2 + (2 - 7)^2 = (5\sqrt{5})^2.$$

6. Solve for c

$$(c - 15)^2 + 25 = 125, \text{ so } (c - 15)^2 = 100 \text{ and } c = 5 \text{ or } 25.$$

7. Find the smaller area

The height from B to $y = 2$ is 5. For $C = (5, 2)$, $AC = 1$, so the minimum area is $\frac{1}{2}(1)(5) = \frac{5}{2}$.

Final answer

(a) $5x - 11y + 2 = 0$.

(b) $C = (5, 2)$ or $(25, 2)$.

(c) $\frac{5}{2}$.

Question 10

Integration

10. The curve C has equation $y = f(x)$ where $x > 0$

Given that

- $f'(x) = 6x - \frac{(2x-1)(3x+2)}{2\sqrt{x}}$

- the point $P(4, 12)$ lies on C

(a) find the equation of the normal to C at P , giving your answer in the form $y = mx + c$ where m and c are integers to be found,

(4)

(b) find $f(x)$, giving each term in simplest form.

(6)

(Total for Question 10 is 10 marks)

Worked Solution - Question 10

1. Expand the numerator

$$(2x - 1)(3x + 2) = 6x^2 + x - 2.$$

2. Simplify f prime

$$f'(x) = 6x - \frac{6x^2 + x - 2}{2\sqrt{x}} = 6x - 3x^{3/2} - \frac{1}{2}x^{1/2} + x^{-1/2}.$$

3. Find the tangent gradient at P

$$f'(4) = 24 - 3(8) - \frac{1}{2}(2) + \frac{1}{2} = -\frac{1}{2}.$$

4. Use the normal gradient

The normal gradient is the negative reciprocal of $-\frac{1}{2}$, so it is 2.

5. Find the normal equation

Through $P(4, 12)$, $y - 12 = 2(x - 4)$, hence $y = 2x + 4$.

6. Integrate f prime

$$f(x) = 3x^2 - \frac{6}{5}x^{5/2} - \frac{1}{3}x^{3/2} + 2x^{1/2} + c.$$

7. Use P to find c

$$12 = 3(4)^2 - \frac{6}{5}(4^{5/2}) - \frac{1}{3}(4^{3/2}) + 2(4^{1/2}) + c.$$

8. Simplify c

$$12 = 48 - \frac{192}{5} - \frac{8}{3} + 4 + c = \frac{164}{15} + c, \text{ so } c = \frac{16}{15}.$$

Final answer

(a) $y = 2x + 4$.

(b) $f(x) = 3x^2 - \frac{6}{5}x^{5/2} - \frac{1}{3}x^{3/2} + 2x^{1/2} + \frac{16}{15}$.

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6 marks

Question 11

Trigonometric Functions

11.

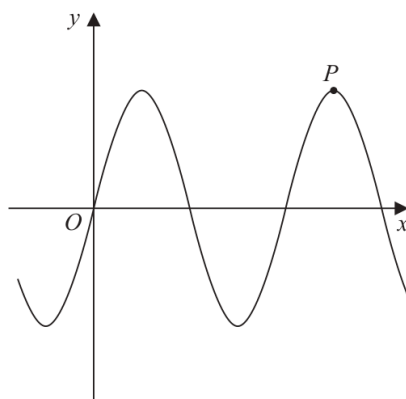


Figure 4

Figure 4 shows a sketch of part of the curve C_1 with equation

$$y = 12 \sin x$$

where x is measured in radians.

The point P shown in Figure 4 is a maximum point on C_1

(a) Find the coordinates of P .

(2)

The curve C_2 has equation

$$y = 12 \sin x + k$$

where k is a constant.

Given that the **maximum** value of y on C_2 is 3

(b) find the coordinates of the **minimum** point on C_2 which has the **smallest** positive x coordinate.

(2)

The curve C_3 has equation

$$y = 12 \sin(x + B)$$

where B is a positive constant.

Given that $\left(\frac{\pi}{4}, A\right)$, where A is a constant, is the **minimum** point on C_3 which has the **smallest** positive x coordinate,

(c) find

(i) the value of A ,

(ii) the smallest possible value of B .

(2)

(Total for Question 11 is 6 marks)

TOTAL FOR PAPER IS 75 MARKS

Worked Solution - Question 11

Primary topic

1. Use maxima of sine

For $y = 12 \sin x$, maxima occur when $x = \frac{\pi}{2} + 2\pi n$, with value 12.

2. Read point P

The maximum point P shown is the second positive maximum, so $P = (\frac{5\pi}{2}, 12)$.

3. Find k for C2

For $y = 12 \sin x + k$, the maximum value is $12 + k$. Given the maximum is 3, $k = -9$.

4. Find the smallest positive minimum on C2

The minimum value is $-12 - 9 = -21$, and it first occurs at $x = \frac{3\pi}{2}$. So the point is $(\frac{3\pi}{2}, -21)$.

5. Find A for C3

For $y = 12 \sin(x + B)$, any minimum has value -12 , so $A = -12$.

6. Find the smallest positive B

At the minimum $x = \frac{\pi}{4}$, so $\frac{\pi}{4} + B = \frac{3\pi}{2}$. Hence $B = \frac{5\pi}{4}$.

Final answer

(a) $P = (\frac{5\pi}{2}, 12)$.

(b) $(\frac{3\pi}{2}, -21)$.

(c)(i) $A = -12$.

(c)(ii) $B = \frac{5\pi}{4}$.